

concentration is 140 millimolar (mM) inside the cell, but only 5 mM outside. The chemical concentration gradient thus favors a net outflow of K^+ . Furthermore, a resting neuron has many open potassium channels, but very few open sodium channels. Because Na^+ and other ions can't readily cross the membrane, K^+ outflow leads to a net negative charge inside the cell. This buildup of negative charge within the neuron is the major source of the membrane potential.

What stops the buildup of negative charge? The excess negative charges inside the cell exert an attractive force that opposes the flow of additional positively charged potassium ions out of the cell. The separation of charge (voltage) thus results in an electrical gradient that counterbalances the chemical concentration gradient of K^+ .

Modeling the Resting Potential

The net flow of K^+ out of a neuron proceeds until the chemical and electrical forces are in balance. We can model this process by considering a pair of chambers separated by an artificial membrane. To begin, imagine that the membrane contains many open ion channels, all of which allow only K^+ to diffuse across (**Figure 48.8a**). To produce a K^+ concentration gradient like that of a mammalian neuron, we place a solution of 140 mM potassium chloride (KCl) in the inner chamber and 5 mM KCl in the outer chamber. The K^+ will diffuse down its concentration gradient into the outer chamber. But because the chloride ions (Cl^-) lack a means of crossing the membrane, there will be an excess of negative charge in the inner chamber.

When our model neuron reaches equilibrium, the electrical gradient will exactly balance the chemical gradient so that no further net diffusion of K^+ occurs across the membrane.

The magnitude of the membrane voltage at equilibrium for a particular ion is called that ion's **equilibrium potential** (E_{ion}). For a membrane permeable to a single type of ion, E_{ion} can be calculated using a formula called the *Nernst equation*. At human body temperature (37°C) and for an ion with a net charge of 1+, such as K^+ or Na^+ , the Nernst equation is

$$E_{ion} = 62 \text{ mV} \left(\log \frac{[ion]_{outside}}{[ion]_{inside}} \right)$$

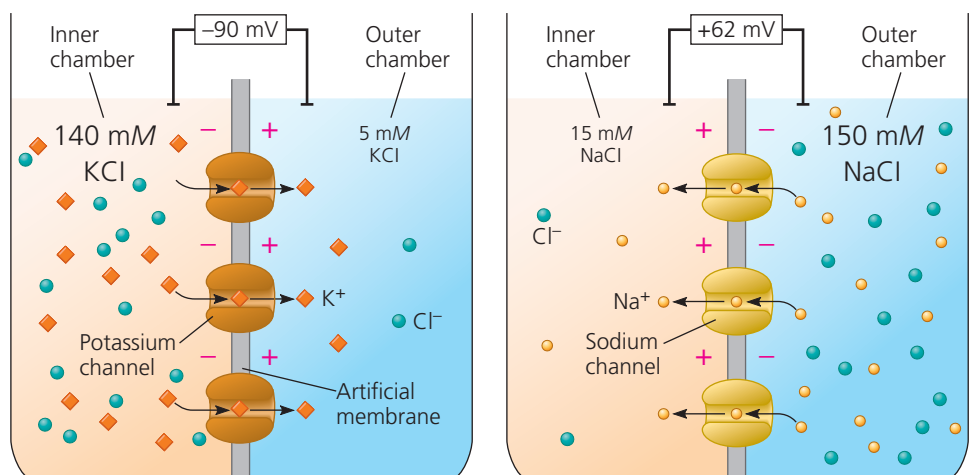
Plugging the K^+ concentrations into the Nernst equation reveals that the equilibrium potential for K^+ (E_K) is -90 mV (see **Figure 48.8a**). The minus sign indicates that K^+ is at equilibrium when the inside of the membrane is 90 mV more negative than the outside.

Whereas the equilibrium potential for K^+ is -90 mV , the resting potential of a mammalian neuron is somewhat less negative. This difference reflects the small but steady movement of Na^+ across the few open sodium channels in a resting neuron. The concentration gradient of Na^+ has a direction opposite to that of K^+ (see **Table 48.1**). Na^+ therefore diffuses into the cell, making the inside of the cell less negative. If we model a membrane in which the only open channels are selectively permeable to Na^+ , we find that a tenfold higher concentration of Na^+ in the outer chamber results in an equilibrium potential (E_{Na}) of $+62 \text{ mV}$ (**Figure 48.8b**). In an actual neuron, the resting potential (-60 to -80 mV) is much closer to E_K than to E_{Na} because there are many open potassium channels but only a small number of open sodium channels.

Because neither K^+ nor Na^+ is at equilibrium in a resting neuron, there is a net flow of each ion across the membrane. The resting potential remains steady, which means that these K^+ and Na^+ currents are equal and opposite. Ion

► **Figure 48.8 Modeling a mammalian neuron.** In this model of the membrane potential of a resting neuron, an artificial membrane divides each container into two chambers. Ion channels allow free diffusion for particular ions, resulting in the net ion flow represented by arrows. **(a)** The presence of open potassium channels makes the membrane selectively permeable to K^+ , and the inner chamber contains a 28-fold higher concentration of K^+ than the outer chamber; at equilibrium, the inside of the membrane is -90 mV relative to the outside. **(b)** The membrane is selectively permeable to Na^+ , and the inner chamber contains a tenfold lower concentration of Na^+ than the outer chamber; at equilibrium, the inside of the membrane is $+62 \text{ mV}$ relative to the outside.

WHAT IF? How would adding potassium or chloride channels to the membrane in (b) affect the membrane potential?



(a) Membrane selectively permeable to K^+

Nernst equation for K^+ equilibrium potential at 37°C:

$$E_K = 62 \text{ mV} \left(\log \frac{5 \text{ mM}}{140 \text{ mM}} \right) = -90 \text{ mV}$$

(b) Membrane selectively permeable to Na^+

Nernst equation for Na^+ equilibrium potential at 37°C:

$$E_{Na} = 62 \text{ mV} \left(\log \frac{150 \text{ mM}}{15 \text{ mM}} \right) = +62 \text{ mV}$$